MEMS 3430: Design Homework #1
Brayton Topping Cycle - Marine Propulsion

February 9, 2026

1 Assumptions and Boundary Conditions

The marine propulsion thermodynamic cycle uses the below assumptions and boundary conditions to analyze the problem:

- **Environmental Temperature (T_1):** 300 K.
- **Maximum Material Temperature (T_3):** 1200 K.
- **Heat Transfer Exit Temperature (T_5):** 500 K.
- **Heat Exchanger Load:** 30 MW of thermal energy used to get to the bottoming cycle.
- **Compression Ratio Range (r):** Equation sheet has values range 5 and 30.
- **Working Fluid:** Model air as an ideal gas with standard air properties.
- **Pressure Drops:** Assume $P_2 = P_3$ for the heater and $P_4 = P_5$ for the heat exchanger during the ideal cycle.
- **Ambient Exhaust:** $P_5 = P_1$ where the fluids exhausts into the atmosphere.
- **Efficiencies:** Isentropic efficiencies for the turbine and compressor are assumed as $\eta_t = 0.90$ and $\eta_c = 0.85$ respectively. Determined using typical marine gas turbine efficiencies.

1.1 Marine Propulsion Choices

- **Fuel Choice:** Used Marine Gas Oil for the CCGT system.
- **Heating Value:** The fuel heating value is assumed to be 42,700 kJ/kg.
- **Fouling Factor:** In Table 11.1 given from equations, the representative fouling factor for the fuel is $R'' = 0.0009\ \mathrm{m^2 \cdot K/W}$.
- **Fuel Cost:** The current market price for MGO is around \$769 per metric ton; a calculated MGO cost of \$0.769 per kg.

1.1.1 Environmental Impacts of Ship Fuel Combustion

The use of MGO in marine propulsion systems can have serious environmental impacts. When deciding on an MGO to use, it is important to adhere to environmental standards to preserve wildlife and keep pollution from close populations. Below are important aspects to consider when choosing and MGO [4]:

1. **Sulfur Oxide Emissions:** When a fuel is burned that contains sulfur oxide, there are SO_x emissions. SO_x emissions contribute to soil acidification, acid rain, and results in adverse affects on human and wildlife health [5].

2. **Nitrogen Oxide Emissions:** Combustion of the fuel creates the formation of NO_x, which directly impacts the CO levels within the atmosphere. Higher nitrogen oxide emissions leads to acid rain and destroys marine environments [6].

3. **Particulate Matter:** Ship exhaust results in particulate matter that lower air quality. Particulate matter leads to lung and heart health risks such as heart/lung disease, asthma, respiratory irritation, and more. Environmentally, particulate matter creates a haze that creates and acidic environment and destroys nutrient values on land and in aquatic environments [3].

2 Schematics and Diagrams

Below are diagrams that show the thermodynamic processes of the topping cycle. The states are labeled across the diagram to follow the process of the cycle.

- **T-s Diagram:** Figure 1 illustrates the diagram for the temperature-entropy relationship during the thermodynamic cycle. There is a non-ideal compression ($1 \rightarrow 2$) and ($3 \rightarrow 4$).

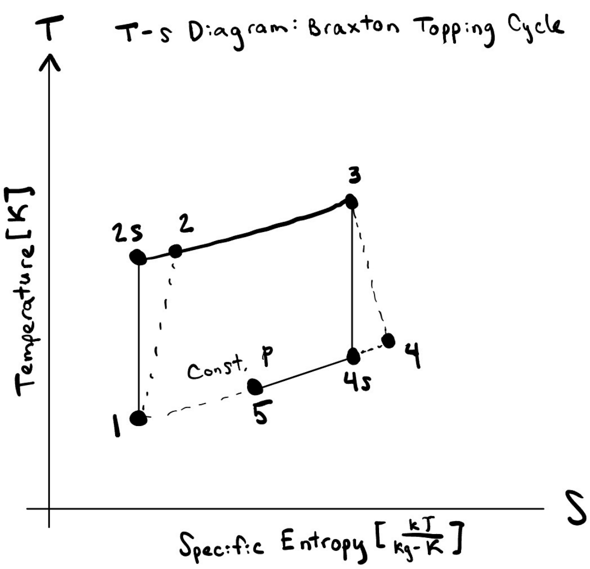

Figure 1: T-s Diagram of the Brayton Topping Cycle

- **p-v Diagram:** Figure 2 illustrates the diagram for the pressure-volume relationships during the thermodynamic cycle. There is a constant pressure heat addition from process (2 → 3) and a heat rejection process to the bottoming cycle from steps (4 → 5).

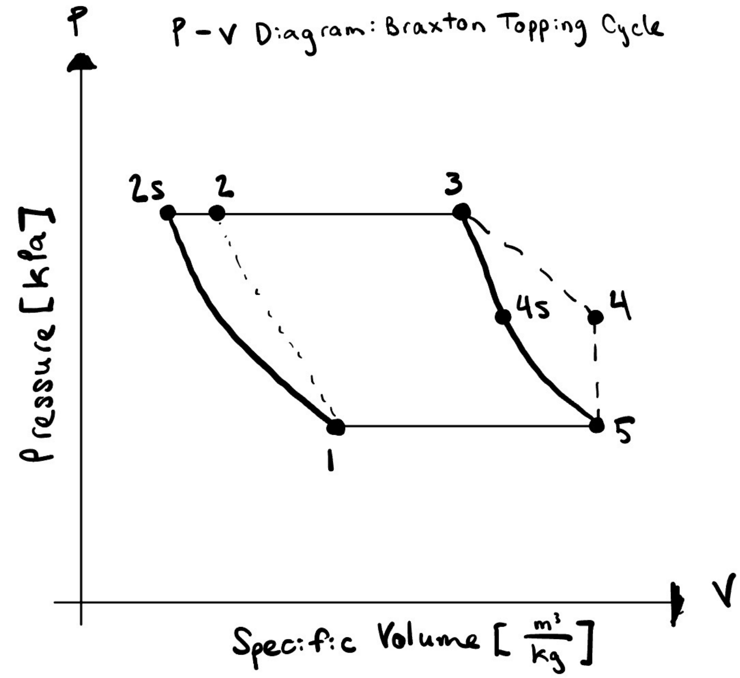

Figure 2: p-v Diagram of the Brayton Topping Cycle.

Within the diagram, the dotted lines represent the ideal states. It is important to note that where there are no pressure drops in the T-s Diagram: (P2 = P3 and P4 = P5). During the p-v Diagram, note that P5 = P1. The diagrams are made using the assumptions and boundary conditions stated previously.

3 Outline of Main Equations

The below equations are found using course materials and the provided equation sheet. These equations will be used to understand the performance of the thermodynamic cycle [2].

3.1 Cycle Thermodynamics and Pressure Relations

The cycle pressures are governed by the compression ratio r and the assumption of constant pressure heat addition and rejection:

$$P_2 = P_1 \cdot r \quad \text{and} \quad P_3 = P_2 \tag{1}$$

$$P_4 = P_1 \quad \text{and} \quad P_5 = P_1 \tag{2}$$

Using the prescribed boundary conditions and isentropic efficiencies η_c and η_t, the state temperatures are determined as follows:

$$T_{2s} = T_1(r)^{\frac{k-1}{k}} \quad \text{and} \quad T_2 = T_1 + \frac{T_{2s} - T_1}{\eta_c} \tag{3}$$

$$T_{4s} = T_3 \left(\frac{1}{r}\right)^{\frac{k-1}{k}} \quad \text{and} \quad T_4 = T_3 - \eta_t(T_3 - T_{4s}) \tag{4}$$

3.2 Performance and Energy Balance

The mass flow rate is derived from the heat rejection to the bottoming cycle $\dot{Q}_{out}$:

$$\dot{m} = \frac{\dot{Q}_{out}}{c_p(T_4 - T_5)} \tag{5}$$

The net power output ($\dot{W}_{net}$) and the thermal efficiency (η_{th}) of the topping cycle:

$$\dot{W}_{net} = \dot{m}[c_p(T_3 - T_4) - c_p(T_2 - T_1)] \tag{6}$$

$$\eta_{th} = \frac{\dot{W}_{net}}{\dot{Q}_{in}} = \frac{\dot{W}_{net}}{\dot{m}c_p(T_3 - T_2)} \tag{7}$$

3.3 Heat Transfer and Fluid Dynamics

For turbulent flow, the Nusselt number is used to determine the internal convection for heat exchangers:

$$Re = \frac{4\dot{m}}{\pi D \mu} \quad \text{and} \quad Nu_D = 0.023 Re_D^{4/5} Pr^n \tag{8}$$

Log Mean Temperature Difference used for the sizing of the heat exchanger:

$$q = UA\Delta T_{lm} \quad \text{where} \quad \Delta T_{lm} = \frac{\Delta T_2 - \Delta T_1}{\ln(\Delta T_2/\Delta T_1)} \tag{9}$$

Pressure drop (Δp) and pump power (P_p):

$$\Delta p = f\rho \frac{u_m^2}{2} \frac{L}{D} \quad \text{and} \quad P_p = \frac{\dot{m}}{\rho}\Delta p \tag{10}$$

3.4 Mass Flow and Fuel Consumption Derivations

Mass flow rate with be used to find the size and cost of the subject propulsion system. The mass flow rate, $\dot{m}$, is derived using a 30 MW heat load ($\dot{Q}_{out}$) that is rejected to the bottoming cycle:

$$\dot{m} = \frac{\dot{Q}_{out}}{c_p(T_4 - T_5)} \tag{11}$$

The heat input ($\dot{Q}_{in}$), which is used to determine the fuel mass flow rate, is calculated as:

$$\dot{Q}_{in} = \dot{m} c_p (T_3 - T_2) \tag{12}$$

The fuel mass flow rate ($\dot{m}_{fuel}$) used to get the specific heat is found using the Lower Heating Value of the fuel:

$$\dot{m}_{fuel} = \frac{\dot{Q}_{in}}{LHV} \tag{13}$$

The LHV used for the MGO is 42,700 kJ/kg; this was stated in the assumptions.

To find the hourly cost for the vessel operation, use the market cost of fuel (C_{fuel}):

$$\text{Cost per Hour } (\$/h) = \dot{m}_{fuel} \cdot 3600 \cdot C_{fuel} \tag{14}$$

The cost per hour converts the cost from seconds to hours. The C_{fuel} is \$0.769/kg.

4 Results and Discussion

4.1 Part A: Parametric Optimization

Analyzing thermodynamic cycle performance for the combinations isentropic efficiencies. Below, table 1 displays the isentropic efficiencies and their according optimal compression ratio.

Table 1: Optimal Compression Ratios for Differing Isentropic Efficiencies

$[\eta_t, \eta_c]$	Optimal r	Max η_{th}
$[1.0, 1.0]$	21.4	58.32%
$[1.0, 0.9]$	21.4	54.34%
$[0.9, 0.9]$	18	37.95%
$[0.8, 0.8]$	7.8	21.11%

4.1.1 Performance Plots

Figure 3 and Figure 4 display the cycle behavior. Different colors are used to signify the various combinations of efficiency.

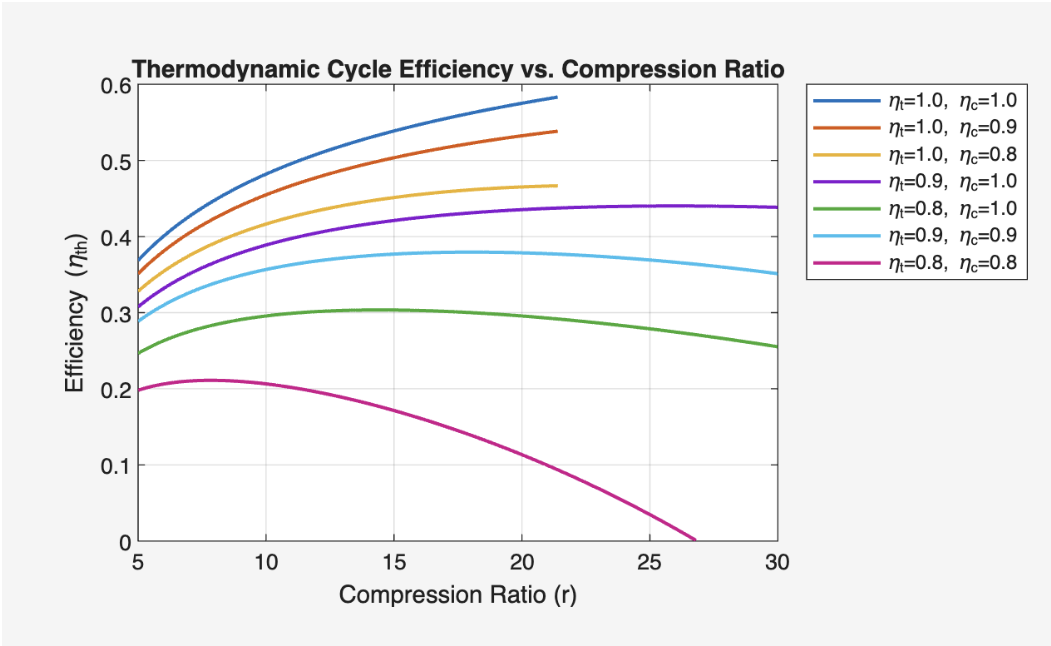

Figure 3: Plot of thermodynamic cycle efficiency vs. compression ratio different isentropic efficiency combinations. Made using Gemini and MATLAB.

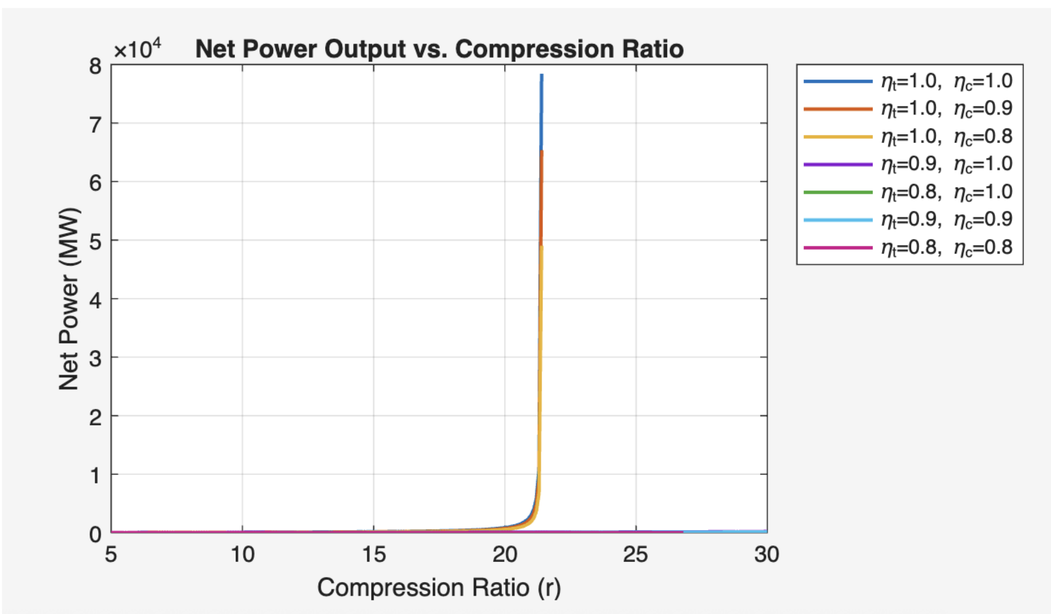

Figure 4: Plot of net power output vs. compression ratio. Made using Gemini and MATLAB.

Discontinuity Discussion: When looking at the lower isentropic efficiencies according to the plots, negative values come across for the net power at high compression ratios. These negative values represent a discontinuity. Since the values represent a discontinuity, it means that the turbine cannot create enough work that is needed for the compressor to function. Therefore, the cycle cannot be completed using efficiencies that have negative values.

Recommendation: Using the understanding of the plots, the turbine efficiencies should be prioritized when determining thermal efficiencies because turbine efficiencies allows users to determine for valuable real-world applications.

4.2 Part B: Final Marine Propulsion Results

Using realistic efficiencies ($\eta_t = 0.90, \eta_c = 0.85$), the following final parameters were determined for the propulsion system:

- **Optimal Compression Ratio (r):** 7.80
- **Max Thermodynamic Efficiency:** 21.11 %
- **Mass Flow Rate ($\dot{m}$):** 135.35 kg/s
- **Net Power Output:** 26.89 MW
- **Required Heat Input:** 84.10 MW
- **Fuel Consumption:** 1.9695 kg/s
- **Hourly Operational Cost:** $5452.31/h

4.3 Sanity Check

The calculated thermal efficiency of the topping cycle is 21.11%. When comparing the maximum thermal efficiency to combined cycle gas turbine systems, this seems a bit low. The thermal efficiency may be low since it only accounts for the topping cycle. Given the assumptions and boundary conditions for this specific system, the thermal efficiency gives significant room for the bottoming cycle which makes sense intuitively. Even though the topping efficiency is not up to current standards, it ensures that the thermal energy is available for an efficient bottoming cycles. The cost of $5452.31/h represent a reliable cost for vessel trips in accordance with current prices.

5 Bibliography

1 Washington University in St. Louis. "DHW 1: Brayton topping cycle," 2026.

2 Washington University in St. Louis. "Heat Transfer Equations Handout," 2026.

3 U.S. Environmental Protection Agency (EPA). "Health and Environmental Effects of Particulate Matter (PM)," 2023. Accessed Feb 2, 2026.

4 U.S. Environmental Protection Agency (EPA). "MARPOL Annex VI and the Act to Prevent Pollution from Ships (APPS): Health and Environmental Effects," 2022. Accessed Feb 2, 2026.

5 California Air Resources Board (CARB). "Sulfur Dioxide and Health," 2024. Accessed Feb 2, 2026.

6 ScienceDirect. "NOx Emissions: An Overview," 2023. Accessed Feb 2, 2026.

7 Washington University in St. Louis. "MEMS 3430 - Design Homework Template," 2026.

8 Google Gemini. "Calculated optimization and fuel cost values for MEMS 3430 DHW 1," Prompt: Can you create matlab code so I can run a simulation for the isentropic efficiencies for the values I give you, 2026.

A MATLAB Script

The following appendix shows the MATLAB code used to find the parametric optimization for Part A and Part B.

```matlab
%% MEMS 3430 - DHW 1: Part A Simulation
clear; clc; close all;

% --- Boundary Conditions & Constants ---
T1 = 300;       % [K] Ambient Inlet Temp
T3 = 1200;      % [K] Turbine Inlet Temp
T5 = 500;       % [K] Heat Exchanger Exit Temp
Q_out_HX = 30e6; % [W] Heat transferred to bottoming cycle (30 MW)

cp = 1005;      % [J/kg-K] Specific heat of air
k = 1.4;        % Isentropic coefficient for air
% Compression ratio range (Primary Optimization Parameter)
r = 5:0.1:30;

% Isentropic Efficiency Combinations [nt, nc]
eta_pairs = [1.0, 1.0; 1.0, 0.9; 1.0, 0.8; 0.9, 1.0; 0.8, 1.0; 0.9, 0.9; 0.8, 0.8];
labels = {'\eta_t=1.0, \eta_c=1.0', '\eta_t=1.0, \eta_c=0.9', ...
          '\eta_t=1.0, \eta_c=0.8', '\eta_t=0.9, \eta_c=1.0', ...
          '\eta_t=0.8, \eta_c=1.0', '\eta_t=0.9, \eta_c=0.9', ...
          '\eta_t=0.8, \eta_c=0.8'};

% Pre-allocate storage for results
eff_matrix = zeros(length(r), size(eta_pairs, 1));
power_matrix = zeros(length(r), size(eta_pairs, 1));

% --- Simulation Loop ---
for i = 1:size(eta_pairs, 1)
    eta_t = eta_pairs(i,1);
    eta_c = eta_pairs(i,2);

    for j = 1:length(r)
        curr_r = r(j);

        % Compressor Exit (State 2)
        T2s = T1 * curr_r^((k-1)/k);
        T2 = T1 + (T2s - T1)/eta_c;

        % Turbine Exit (State 4)
        T4s = T3 * (1/curr_r)^((k-1)/k);
        T4 = T3 - eta_t * (T3 - T4s);

        % Check Physical Feasibility
        % (1) T4 must be greater than T5 to reject heat to the bottoming cycle
        % (2) T2 must be less than T3 for heat addition in the heater
        if T4 > T5 && T2 < T3
            % Mass flow rate required for 30 MW heat load
            m_dot = Q_out_HX / (cp * (T4 - T5));

            % Specific Work [J/kg]
            w_turbine = cp * (T3 - T4);
            w_compressor = cp * (T2 - T1);

            % Net Power Output [W]
            W_net = m_dot * (w_turbine - w_compressor);

            % Thermal Efficiency
            eta_th = (w_turbine - w_compressor) / (cp * (T3 - T2));

```

Figure 5: Screenshot of the MATLAB used for topping cycle analysis.

```matlab
            % Store only positive net power results
            if W_net > 0
                eff_matrix(j, i) = eta_th;
                power_matrix(j, i) = W_net / 1e6; % Convert to MW
            else
                eff_matrix(j, i) = NaN;
                power_matrix(j, i) = NaN;
            end
        else
            eff_matrix(j, i) = NaN;
            power_matrix(j, i) = NaN;
        end
    end
end

% --- Plotting Results ---
figure;
subplot(2,1,1);
plot(r, eff_matrix, 'LineWidth', 1.5);
title('Thermodynamic Cycle Efficiency vs. Compression Ratio');
xlabel('Compression Ratio (r)'); ylabel('Efficiency (\eta_{th})');
legend(labels, 'Location', 'bestoutside'); grid on;

subplot(2,1,2);
plot(r, power_matrix, 'LineWidth', 1.5);
title('Net Power Output vs. Compression Ratio');
xlabel('Compression Ratio (r)'); ylabel('Net Power (MW)');
legend(labels, 'Location', 'bestoutside'); grid on;

% --- Find Optima for Table 4.1 ---
fprintf('--- Optimization Results (Section 4.1) ---\n');
for i = 1:size(eta_pairs, 1)
    [max_eff, idx] = max(eff_matrix(:,i));
    if ~isnan(max_eff)
        fprintf('%s: Opt r = %.1f, Max Efficiency = %.2f%%\n', ...
            labels{i}, r(idx), max_eff*100);
    else
        fprintf('%s: No physically feasible cycle found.\n', labels{i});
    end
end
```

Figure 6: Screenshot of the MATLAB used for topping cycle analysis.

```
%% MEMS 3430 - DHW 1: Topping Cycle Solver
clear; clc; close all;

% --- Boundary Conditions (Given) ---
T1 = 300;       % [K] Ambient Temp
T3 = 1200;      % [K] Turbine Inlet Temp
T5 = 500;       % [K] Exhaust to Bottoming Cycle
Q_out = 30e6;   % [W] Heat to Bottoming Cycle
P1 = 101325;    % [Pa] Ambient Pressure

% --- Constants ---
cp = 1005;      % [J/kg-K] Specific heat of air
k = 1.4;        % Isentropic coefficient
LHV = 42700e3;  % [J/kg] Fuel heating value (MGO)
fuel_cost_kg = 0.769; % [$/kg] Based on $769/metric ton

% Compression ratio range
r_range = 5:0.1:30;

% Efficiency combinations for Part A
eta_pairs = [1.0, 1.0; 1.0, 0.9; 1.0, 0.8; 0.9, 1.0; 0.8, 1.0; 0.9, 0.9; 0.8, 0.8];
labels = {'100/100', '100/90', '100/80', '90/100', '80/100', '90/90', '80/80'};

figure(1); hold on; grid on; title('Part A: Cycle Efficiency vs. r');
figure(2); hold on; grid on; title('Part A: Net Power vs. r');

for i = 1:size(eta_pairs, 1)
    etat = eta_pairs(i,1);
    etac = eta_pairs(i,2);

    eta_th = zeros(size(r_range));
    W_net_spec = zeros(size(r_range));

    for j = 1:length(r_range)
        r = r_range(j);

        % State 2 (Compressor)
        T2s = T1 * r^((k-1)/k);
        T2 = T1 + (T2s - T1)/etac;

        % State 4 (Turbine)
        T4s = T3 * (1/r)^((k-1)/k);
        T4 = T3 - etat * (T3 - T4s);

        % Check for Physicality
        w_comp = cp * (T2 - T1);
        w_turb = cp * (T3 - T4);

        if T2 < T3 && w_turb > w_comp
            W_net_spec(j) = w_turb - w_comp;
            eta_th(j) = W_net_spec(j) / (cp * (T3 - T2));
        else
            W_net_spec(j) = NaN; % Discontinuity/Infeasible
            eta_th(j) = NaN;
        end
    end

```

Figure 7: Screenshot of the MATLAB used for topping cycle analysis.

```matlab
%% --- Part B: Realistic Case (Marine Propulsion) ---
etat_real = 0.90; % Realistic values
etac_real = 0.85;

[~, idx] = max(eta_th); % Finding optimum r
opt_r = r_range(idx);

% Final Calculations
T2s_f = T1 * opt_r^((k-1)/k);
T2_f = T1 + (T2s_f - T1)/etac_real;
T4s_f = T3 * (1/opt_r)^((k-1)/k);
T4_f = T3 - etat_real * (T3 - T4s_f);

m_dot = Q_out / (cp * (T4_f - T5)); % Cycle mass flow
W_net = m_dot * cp * ((T3 - T4_f) - (T2_f - T1)); % Net Power
Q_in = m_dot * cp * (T3 - T2_f); % Heat Input
m_fuel = Q_in / LHV; % Fuel mass flow
hourly_cost = m_fuel * 3600 * fuel_cost_kg; % Cost

% Displaying Results for Section 4.2
fprintf('--- Part B: Final Answers ---\n');
fprintf('Optimal r: %.2f\n', opt_r);
fprintf('Max Efficiency: %.2f%%\n', max(eta_th)*100);
fprintf('Mass Flow Rate: %.2f kg/s\n', m_dot);
fprintf('Net Power Output: %.2f MW\n', W_net/1e6);
fprintf('Required Heat Input: %.2f MW\n', Q_in/1e6);
fprintf('Fuel Consumption: %.4f kg/s\n', m_fuel);
fprintf('Hourly Operational Cost: $%.2f/h\n', hourly_cost);
```

Figure 8: Screenshot of the MATLAB used for topping cycle analysis.